

GEOLOGIC HAZARD

Landslide Zones

A slope can move slowly for years or fail within seconds after rain, earthquake shaking, fire, erosion, or grading.



1 Understand the factor

What it is - and why it matters

Landslides include slow soil creep, rotational slides, rockfalls, debris slides, and fast debris flows. Hazard depends on slope angle, geology, old slide deposits, groundwater, vegetation, drainage, wildfire history, grading, retaining systems, and what lies uphill as well as downhill. A mapped zone identifies susceptibility or past movement; it does not establish the exact boundary, depth, speed, or probability of a future slide on a particular parcel.

How risk can reach a household

- Slope material can strike or bury a home, undermine a foundation, or push walls and retaining structures inward.
- Runoff and groundwater can raise pore pressure, reduce soil strength, and reactivate an old landslide.
- Rockfall or debris from an uphill parcel can cross property lines, roads, drainage channels, and utility corridors.
- A downhill failure can remove support beneath a yard, deck, pool, driveway, road, or building pad.

What people often miss

- The most important source area may be above the property, outside the parcel boundary or subdivision.
- Old landslide deposits can look stable and vegetated but may reactivate when drainage or loading changes.
- Wildfire can remove vegetation and alter soil, increasing debris-flow risk during later storms.
- Retaining walls require drainage and maintenance; a wall alone does not prove the slope has been engineered safely.

? Five checks before buying

1

Review official landslide, debris-flow, geology, slope, and wildfire maps for the parcel and surrounding drainage basin.

2

Obtain grading plans, geotechnical reports, retaining-wall permits, drainage easements, and prior movement or repair records.

3

Look for tilted trees or fences, hummocky ground, bulging walls, new cracks, seepage, patched pavement, and offset drains.

4

Have a geotechnical professional evaluate significant slopes, fills, old slide deposits, retaining systems, and drainage.

5

Ask the insurer in writing how earth movement, mudflow, mudslide, debris flow, flooding, and wildfire-related damage are defined.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 03

H Health impacts

- Rapid slides can cause burial, crushing, blunt trauma, drowning, and delayed rescue when roads are blocked.
- Dust, mold, dampness, and contaminated sediment may affect residents during cleanup and reconstruction.
- Broken gas, electrical, water, and sewer systems can add fire, contamination, and sanitation hazards.
- Ongoing slope movement and storm warnings can create chronic anxiety, sleep disruption, and evacuation stress.

P Property impacts

- Movement can crack or rotate foundations, walls, pools, decks, utilities, driveways, roads, and retaining structures.
- Access and utility corridors can be lost even when the home itself is outside the slide body.
- Stabilization may require drainage galleries, soldier piles, tiebacks, soil nails, grading, buttresses, or property acquisition.
- Known movement or geotechnical restrictions can affect appraisals, disclosures, buildable area, permits, financing, and resale.

I Insurance implications

- Most homeowners policies exclude earth movement such as landslide, mudslide, settling, sinking, and soil expansion.
- Specialty difference-in-conditions or earth-movement coverage may exist in some markets but can be limited and expensive.
- Coverage for mudflow or water-carried debris can depend on the policy definition and the event's initiating cause.
- Slope maintenance, drainage correction, stabilization, and loss of land value are commonly owner expenses.

Possible long-term health implications of buying nearby

- The dominant long-term concern is not routine exposure but the possibility of injury, isolation, and displacement during slope failure.
- Moisture intrusion from hillside seepage can support mold and indoor dampness even without a major landslide.
- Repeated movement, repair uncertainty, litigation, and storm evacuations can produce sustained psychological and financial stress.
- Risk can change after wildfire, excavation, irrigation, leaks, or neighboring construction; historical stability is not a permanent guarantee.

Evidence lens

Landslide mechanics and triggers are well established, but mapped susceptibility is not a parcel-specific prediction. Qualified geotechnical evaluation is important for steep or previously moving sites.

Plan more carefully for

Children

Older adults

Pregnancy

Heart/lung conditions

Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 03

V Verify the actual risk

- Inspect the entire slope system, including uphill source areas, downhill support, drainage paths, and adjacent construction.
- Keep geotechnical reports, grading plans, wall designs, maintenance schedules, and photographs of baseline conditions.

P Protect the property

- Maintain roof drains, swales, subdrains, retaining-wall outlets, and erosion protection; repair leaks promptly.
- Do not add fill, pools, heavy structures, irrigation, or major landscaping on slopes without professional review.

M Monitor and respond

- Watch for new cracks, bulges, leaning features, seepage, blocked drains, falling rocks, or sudden changes after storms.
- Follow evacuation warnings and have alternate routes, go-bags, medications, and plans for pets and mobility needs.

I Prepare insurance records

- Request written explanations of earth-movement and water exclusions and any specialty coverage available.
- Document maintenance and pre-loss condition; emergency stabilization may require rapid, uninsured spending.

REAL-LIFE EXAMPLE

2014 Oso Landslide

March 22, 2014

A large landslide near Oso, Washington, crossed the North Fork Stillaguamish River and struck a residential community. The USGS reports 43 fatalities, destruction of about 40 homes and other structures, and burial of a section of State Route 530.

Why it matters: The disaster showed how a fast-moving slide can travel beyond an obvious steep slope, block a river and highway, and overwhelm warning, access, and rescue systems.

[Open official incident record - usgs.gov >](#)

Questions to resolve before making a decision

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2. Obtain grading plans, geotechnical reports, retaining-wall permits, drainage easements, and prior movement or repair records.
3. Look for tilted trees or fences, hummocky ground, bulging walls, new cracks, seepage, patched pavement, and offset drains.
4. Have a geotechnical professional evaluate significant slopes, fills, old slide deposits, retaining systems, and drainage.

Official sources and mapping tools

[USGS Landslide Hazards Program - usgs.gov >](#)

[USGS landslide preparedness - usgs.gov >](#)

[FEMA landslide guidance - ready.gov >](#)